F1 Overtake Window Predictor

DEEP LEARNING • MILESTONE 1 • TEAM JANUSFORGE • 5 NOVEMBER 2026

Team: Owen, Amine, Joshua, Ayobami Idris. **Research question.** Can a feedforward neural network predict whether a Formula 1 driver overtakes the car directly ahead in the **next 60 seconds**, using telemetry and race context from the **previous 30 seconds**? We construct driver-pair periods every 10 seconds.

17,719	665	5
labeled windows	positive windows (3.75%)	2024 F1 races

1. Data and preprocessing

Source. OpenF1 race-session data (https://openf1.org/docs/) for five 2024 Grands Prix: Bahrain, Australia, Spain, Austria, and São Paulo. Raw inputs include driver positions and intervals, laps, overtake events, pit stops, race-control messages, and car telemetry. **Example observation.** Bahrain 2024 at 15:05:42 UTC: car **18** follows car **77**, occupies position **19**, is **1.766 seconds** behind, and does not overtake within the next 60 seconds (label 0). **Preprocessing pipeline.** Reconstruct positions at 10-second intervals and pair adjacent classified positions. Join recent time gaps (at most 20 seconds old); retain 0<gap3 seconds. Exclude nearby pit events (90 seconds), yellow-flag and safety-car/VSC/red-flag periods, and windows extending beyond race finish. We then check overtakes against position changes and sustained order; remove ambiguous or contradictory cases. Verify both drivers have active lap records. **Features.** For each remaining pair, summarize the prior 30 seconds of both cars' speed, throttle, brake usage, RPM, and DRS activity; include gap, position, and relative-speed measures. The final dataset has **16 numeric features**, zero missing values, and no future telemetry in the inputs.

Race-separated dataset split

Partition	Races	Windows	Positive
Train	Bahrain, Spain, Austria	11,095	497
Validation	Australia	2,733	62
Test	São Paulo	3,891	106

All windows from any one race remain in a single partition, avoiding leakage from overlapping snapshots between train and test. Test data was not used to select parameters or decision thresholds.

2. Models and optimization

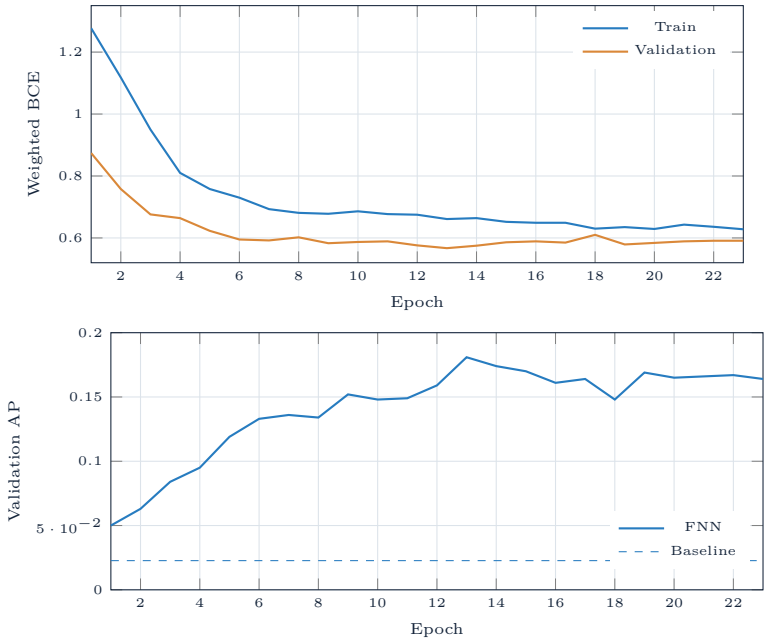
Baseline. Majority class (label 0): always predict *no overtake*; **0 trainable parameters**. **Feedforward Neural Network (FNN).** We used ReLU hidden activations for optimization and dropout 0.15 after the first hidden layer; **1,089 trainable parameters**. StandardScaler is fit to training data only. Training uses "Adam", learning rate 0.001, batch size 256, class-weighted binary cross-entropy (positive weight 21.32), up to 60 epochs and early stopping with patience 10. **Selection and rationale.** The compact network limits capacity for a five-race sample. Class weighting compensates for rare positive windows (4.48% of training data). The checkpoint with the highest Australia validation average precision is from **epoch 13**; the validation-F1 decision threshold is **0.8609**, was frozen before testing.

Training and held-out results

MODEL SELECTION ON AUSTRALIA • FINAL EVALUATION ON SÃO PAULO

3. Optimization evidence: training curves

Curves reconstructed from the experiment's training-plot output for in-document rendering; reported AP and other evaluation metrics below are taken from the actual evaluation.



Training loss declines; validation AP reaches 0.1809 at epoch 13. Early stopping restores the best checkpoint.

4. Error metrics and final test results

Primary metric: average precision (AP, the area summary of the precision–recall curve), appropriate when only 2.72% of test windows are positive. Precision measures correctness among predicted overtakes; recall measures detected actual positives; F1 is their harmonic mean.

Model and set	Accuracy	Precision	Recall	F1	AP (PR-AUC)
Baseline / test	0.9728	0	0	0	0.0272
FNN / validation	—	0.2644	0.3710	0.3087	0.1809
FNN / test	0.9465	0.1645	0.2358	0.1938	0.1148

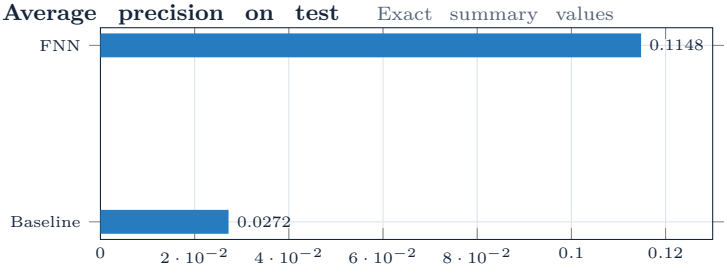
The FNN achieves **4.22 times the baseline test AP**. Its test accuracy is lower because the baseline always predicts the overwhelmingly common negative class, but it cannot identify a single actual overtake.

Test confusion matrix Rows: actual; columns: predicted

Actual	No overtake	Overtake
No overtake	**3,658 TN**	127 FP
Overtake	81 FN	**25 TP**

The model predicts 152 positive windows, of which 25 are correct, and misses 81 of the 106 actual positive windows.

Average precision on test Exact summary values



Error analysis, limitations and next steps

HELD-OUT TEST RACE • SÃO PAULO 2024

5. Error analysis

Gap to car ahead	TP	FN	FP	TN
0–0.5 s	23	13	93	120
0.5–1 s	2	44	34	1,407
1–2 s	0	19	0	1,287
2–3 s	0	5	0	844

Pattern. The model finds **23 of its 25 true positives** when the following car is within 0.5 seconds. But the same band also accounts for **93 of 127 false alarms**. Closing speed and proximity are suggestive, but not sufficient for perfect accuracy. The model misses 44 positive windows at gaps of 0.5–1 seconds and detects none above 1 second, so it is better at identifying imminent passing opportunities than predicting them from farther away. **Concrete errors.** At 16:12:07 UTC in São Paulo, car 27 followed car 14 by 0.466 s with a mean relative speed of +23.82 km/h. Its model score was 0.9976, yet the observed window was negative (false positive). At 16:04:37 UTC, car 1 followed car 30 by 0.468 s; a positive window received a score of 0.8605, just below the validation threshold 0.8609 (false negative). The examples show both false-alarm risk and threshold sensitivity.

Known limitations

- **Limited diversity:** Only five 2024 races; São Paulo involved unusual interruptions and weather.
- **Overlapping windows:** The 665 positive rows are not 665 distinct overtakes; scores are window-level, not maneuver-level.
- **Provisional labels:** Race event and position histories can be incomplete or noisy; uncertainty filters may remove genuine passes.
- **Missing context:** Tyre age/compound, wetness, circuit geometry, traffic density, and gap trends are not included. A seperate dataset would be needed.
- **Uncalibrated scores:** Class weighting helps training, but sigmoid outputs should not be interpreted as verified probabilities.

6. Work planned over the next three weeks

Week 1 – label quality and data: Manually audit recorded passes, add more races, review neutralized intervals, and report event-level metrics. **Week 2 – features and architectures:** Add tyre/stint, weather, gap-rate, sector information and temporal sequences; compare the feedforward network and play with different types of models. **Week 3 – robustness and demonstration:** Conduct multiple held-out-race evaluations, investigate threshold trade-offs, calibrate scores, and create a simple benchmarks to improve the model and iterate quicker.

Reproducibility

Source repository: github.com/amineelmess/f1-overtake-window-predictor-JanusForge . The `milestone1.ipynb` notebook documents preprocessing, the majority-class baseline, FNN training, validation, test metrics, and figures. Dependencies are listed in `requirements.txt`; `preprocess_all_races.py` and `finalize_dataset.py` prepare the reproducible data splits. The notebook and report use the same held-out test race.